Supplementary Material for "Epistemic Bisimulation Geometry: Finite-State Certificates for Representation and Exploration"

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This document contains the appendices of the main paper. Appendix A collects the proofs deferred from the main text. Appendices B–E give the supporting arguments in the order in which the main results use them: the confidence event, the transport estimates that make the operators work on nonmetric costs, the closed loop, and the lower bounds. Appendix F specifies the neural instantiation, Appendix G records the experimental and numerical specifications of the eleven studies, and Appendix H analyzes transport error and computation in the neural extension. Numbers carrying a letter prefix refer to this document; section, equation, theorem, figure, and table numbers without a prefix refer to the main paper, and citations refer to the reference list at the end of this document. The code, specifications, and seed-level records are available at https://github.com/Adityasikarwar3107/epistemic-bisimulation-geometry.

APPENDIX A
PROOFS DEFERRED FROM THE MAIN TEXT

This appendix collects, in the order of the main text, the proofs of the results stated in Sections III–VI whose arguments are short enough to give in full but not needed to follow the exposition.

Proof of Theorem 5. Monotonicity and contraction follow from Lemma 4, since positive parts, maxima, and clipping are nonexpansive. On the confidence event, the true reward difference lies between $(|\hat{r} - \hat{r}'| - e^r_a)_+$ and $|\hat{r} - \hat{r}'| + e^r_a$, and (15) gives the corresponding bounds for the transport term. Hence $\mathcal{T}^-[h] \le \mathcal{T}[h] \le \mathcal{T}^+[h]$ for every $h \in \mathfrak{C}$. Iterating the outer operators from $d_\sim$ and using monotonicity and contraction gives the sandwich. That d^+ is a pseudometric is proved in Appendix C-B. □

Proof of Theorem 6. Dropping the clipping gives $d^+ \le \max_a(q_a(d^+) + b_a)$, and the two positive parts in $\mathcal{T}^-$ give $d^- \ge \max_a(q_a(d^-) - b_a)$. Subtracting and using $\max f - \max g \le \max(f - g)$ reduces the difference to a transport term. Since $\Lambda^a_{s,s'}$ is feasible for the upper cost,

$$\mathrm{OT}_{d^+}(\hat{P}, \hat{P}') - \mathrm{OT}_{d^-}(\hat{P}, \hat{P}') \le \langle \Lambda^a_{s,s'}, \Delta \rangle, \tag{A.1}$$

which proves (20). Because $\mathcal{F}$ is monotone and contractive, $\Delta \le U$. Taking supremum norms gives (21), and the midpoint bound follows from the sandwich. □

Proof of Proposition 7. Clipping is monotone and nonexpansive. Since $\Delta \le R_{\max}$ and $\Delta \le \mathcal{F}[\Delta]$, iterating $\mathcal{F}_c$ from Δ produces an increasing sequence that converges to U_c. Iterating from zero stays below both U and $R_{\max}$, which gives the remaining inequalities. The diagonal pair is absorbing with zero source. □

Proof of Proposition 9. A feasible pair coupling leaves $K \times K$ with probability at most the sum of the two marginal escape probabilities, hence at most $2\eta_K$. In (20), bound the width inside $K \times K$ by its maximum and outside by $R_{\max}$, and rearrange. □

Proof of Proposition 14. For midpoint fitting, an encoder within $\kappa_\Phi + \zeta$ of the true metric is within $\kappa_\Phi + \zeta + \|\Delta\|_\infty / 2$ of the midpoint; apply (36) and let $\zeta \downarrow 0$. For interval fitting, the same encoder has loss at most $\kappa_\Phi + \zeta$, because the true metric lies in every interval. A fitted distance outside an interval differs from any point inside it by at most its interval loss plus the interval width. □

Proof of Lemma 16. For any unresolved row (g, a), some true policy reaches g within $H - 1$ steps with probability at least p and then executes a. Stopping earlier at another unresolved row yields payoff at least $1/2$, so the optimal true expected terminal payoff is at least $p/2$. For any fixed stopped policy, coupling the true and empirical transitions on its known rows shows that its true and empirical expected payoffs differ by at most $H\alpha_*$. Empirical optimization loses this amount once when the true optimizer is evaluated and once more when the empirical optimizer is executed, so the executed policy has true expected payoff at least $p/2 - 2H\alpha_* \ge p/4$. Terminal payoffs are at most one, hence the probability of executing an unresolved row is at least this payoff. □

Proof of Theorem 22. Lift d_x to a cost that depends only on (x, y). Every coupling of the full successor laws projects to an endogenous coupling of the same cost, which gives a lower bound by endogenous transport. Conversely, an optimal endogenous coupling lifts to a full coupling by coupling the exogenous marginals independently, which gives the reverse bound. The lifted cost is therefore a fixed point of (3), and uniqueness proves the claim. □

APPENDIX B
COMPLETE PROBABILITY AND OPERATOR ARGUMENTS

A. Anytime Confidence Proof

Fix a state-action row and imagine an infinite stream of independent samples from its stationary reward-successor law. Each time the agent visits the row, it consumes the next sample of the stream. This construction has the same law as adaptive interaction with the MDP, so confidence that holds simultaneously for every prefix of every stream also holds at the visit counts the agent happens to reach.

For $n \geq 1$, Hoeffding's inequality gives

$$\mathbb{P}\Big(|\hat{r}_n - r| > R_{\max}\sqrt{\ell_n/(2n)}\Big) \leq 2e^{-\ell_n}. \tag{B.1}$$

For a successor law supported on at most Γ labels, the multinomial deviation inequality gives

$$\mathbb{P}\big(\|\hat{P}_n - P\|_1 > u\big) \leq 2^{\Gamma} e^{-nu^2/2}. \tag{B.2}$$

With $u = 2\sqrt{(\Gamma \log 2 + \ell_n)/(2n)}$ the right-hand side is at most $e^{-\ell_n}$. When a radius is clipped at the largest possible error, the corresponding failure event is empty, and the convention at $n = 0$ is deterministic for the same reason.

There are SA rows. Summing both tails over all rows and all $n \geq 1$,

$$\begin{aligned}\mathbb{P}(\mathcal{E}^c) &\leq SA \sum_{n \geq 1} 3 \cdot \frac{3\delta}{4SA\pi^2 n^2} \\ &= \frac{3\delta}{8} \leq \delta,\end{aligned} \tag{B.3}$$

using $\sum_{n \geq 1} n^{-2} = \pi^2/6$. The constants leave deliberate slack. The number of observed successors cannot replace the fixed bound Γ, because a successor not yet observed can have positive mass.

APPENDIX C
TRANSPORT AND METRIC PRESERVATION

Everything in Section III rests on one observation: transport is stable under perturbations of its cost and of its marginals even when the cost violates the triangle inequality. We prove this first and then use it repeatedly.

A. Primal Transport Stability Proof

For any feasible coupling Λ, $|\langle \Lambda, h - \tilde{h} \rangle| \leq \left\| h - \tilde{h} \right\|_{\infty}$, since Λ has total mass one. Evaluating $\tilde{h}$ at an h-optimal plan gives one inequality, and exchanging the two costs gives the other. The same evaluation argument proves monotonicity.

For the perturbation of marginals, let (X, Y) be distributed according to an optimal coupling of (μ, ν). Conditionally on (X, Y), couple X maximally with $\tilde{X} \sim \tilde{\mu}$ and Y maximally with $\tilde{Y} \sim \tilde{\nu}$. This gives a feasible coupling of $(\tilde{\mu}, \tilde{\nu})$ with

$$\mathbb{P}((X, Y) \neq (\tilde{X}, \tilde{Y})) \leq \mathrm{TV}(\mu, \tilde{\mu}) + \mathrm{TV}(\nu, \tilde{\nu}). \tag{C.1}$$

Both costs take values in $[0, R_{\max}]$, so their expectations under the two couplings differ by at most $R_{\max}$ times this probability. The perturbed optimum is therefore at most the original optimum plus the stated error, and reversing the roles gives the absolute bound. The argument never uses a triangle inequality for h.

B. Full Epistemic Sandwich Proof

Each operator returns a symmetric, nonnegative cost with zero diagonal, bounded by $R_{\max}$. For the lower operator this holds because each positive part is at most the corresponding uncorrected reward or transport term, and the convex combination of those terms is at most $R_{\max}$. Both operators therefore map $\mathfrak{C}$, a closed subset of a finite-dimensional Banach space, into itself. For every pair,

$$|\mathcal{T}^{\pm}[h](s, s') - \mathcal{T}^{\pm}[\tilde{h}](s, s')| \leq \gamma \left\| h - \tilde{h} \right\|_{\infty}, \tag{C.2}$$

so Banach's theorem gives a unique fixed point for each operator.

On $\mathcal{E}$, the triangle inequality for the reward estimates and Lemma 4 give the operator comparison. The true operator takes values in $[0, R_{\max}]$, so clipping the upper operator cannot reverse it. Iterating the upper operator from $d_{\sim}$ gives $d_{\sim} \leq \mathcal{T}^{+}[d_{\sim}] \leq (\mathcal{T}^{+})^2[d_{\sim}] \leq \cdots$, with limit d^{+}; iterating the lower operator from $d_{\sim}$ gives a decreasing sequence with limit d^{-}. This proves the sandwich.

It remains to show that d^{+} satisfies the triangle inequality. For a pseudometric h, transport between probability measures is a pseudometric: glue optimal couplings of (μ, ν) and (ν, ξ) into a coupling of three variables and use $h(X, Z) \leq h(X, Y) + h(Y, Z)$. For each action define

$$c_a(s) = (1 - \gamma)\beta^r(s, a) + \gamma R_{\max}\beta^P(s, a), \tag{C.3}$$

so that the confidence term added off the diagonal is $c_a(s) + c_a(s')$. For three distinct labels, this star cost satisfies the triangle inequality because the two-step sum contains the extra nonnegative term $2c_a(x)$; cases with coinciding labels are immediate. The reward distance, the transport distance, and the star cost are pseudometrics, and they enter with total weight one. A maximum over actions of pseudometrics is a pseudometric. Finally, if g is a pseudometric then so is $\min\{R_{\max}, g\}$: either the two truncated terms on the right sum to at least $R_{\max}$, or the truncation does not affect the ordinary triangle bound. Iterating $\mathcal{T}^{+}$ from the zero pseudometric and passing to the limit shows that d^{+} is a pseudometric.

C. Value Bound and Its Policy Limitation

Let $v^* = (1 - \gamma)V^*$, whose Bellman operator is $Bv(s) = \max_a[(1 - \gamma)r(s, a) + \gamma \mathbb{E}_{P_{s,a}} v]$. If v is 1-Lipschitz in $d_{\sim}$, then every feasible coupling gives $|\mathbb{E}_{\mu} v - \mathbb{E}_{\nu} v| \leq \langle \Lambda, d_{\sim} \rangle$. Minimizing over couplings and comparing the maxima over actions yields $|Bv(s) - Bv(s')| \leq \mathcal{T}[d_{\sim}](s, s') = d_{\sim}(s, s')$. Starting value iteration at zero and passing to the limit proves the bound. The proof for a fixed policy is identical on the induced one-action model.

The bound fails for arbitrary policies with $d_{\sim}$. Take two labels with identical reward vectors $(0, 1)$ that self-loop under both actions, so that $d_{\sim} = 0$. A policy that takes the zero-reward action at one label and the unit-reward action at the other has value difference $1/(1-\gamma)$. The policy's own induced metric does reflect this difference.

D. Pair Majorant and Occupancy Formula

For each action and pair, the lower fixed point is at least $q_a(d^-) - b_a$, even when this expression is negative, and the upper fixed point is at most $\max_a[q_a(d^+) + b_a]$. The elementary inequality for differences of maxima, together with (A.1), proves the pair Bellman inequality. The map $\mathcal{F}$ is monotone and γ-contractive on bounded functions of pairs, and its rewards are bounded and nonnegative, so iterating it from Δ gives an increasing sequence that converges to U.

For a fixed stationary pair policy, summing the geometric series $\sum_{j\geq 0} \gamma^j (P^\sigma_\Lambda)^j$ gives (25). A finite discounted MDP has a stationary deterministic optimal policy whose value is the unique Bellman fixed point, which proves (24). Diagonal plans stay on diagonal pairs and collect zero reward, so $U(s,s) = 0$.

E. Refined Endpoints

Monotonicity and contraction follow as in Appendix C-B, because the constants π_a and $\bar{D}$ do not depend on h. On $\mathcal{E}$, the true reward difference lies in $[\underline{g}_a, \bar{g}_a]$. For the transport term, let $h = d_\sim$, a pseudometric. Take an optimal coupling of $(\hat{P}_{s,a}, \nu)$ and a maximal coupling $(\hat{X}, X)$ of $\hat{P}_{s,a}$ and $P_{s,a}$. Gluing them gives a coupling of $(P_{s,a}, \nu)$, and the triangle inequality $h(X,Y) \leq h(\hat{X},Y) + h(X,\hat{X})$ yields $\mathrm{OT}_h(P_{s,a},\nu) \leq \mathrm{OT}_h(\hat{P}_{s,a},\nu) + D_{s,a}\,\mathrm{TV}(P_{s,a},\hat{P}_{s,a})$, because X and $\hat{X}$ both lie in $\Sigma_{s,a}$. The reverse inequality and the same step for the second marginal give $|\,\mathrm{OT}_{d_\sim}(P_{s,a},P_{s',a}) - \mathrm{OT}_{d_\sim}(\hat{P}_{s,a},\hat{P}_{s',a})| \leq \pi_a/\gamma$. Hence $\mathcal{T}^-_D[d_\sim] \leq \mathcal{T}[d_\sim] = d_\sim \leq \mathcal{T}^+_D[d_\sim]$, where the clip at $\bar{D}$ is harmless because $d_\sim \leq \bar{D}$. Iterating the monotone contractions from $d_\sim$ proves the sandwich. For the pseudometric property, $\bar{g}_a$ with zero diagonal satisfies the triangle inequality, since $\bar{r}_s - \underline{r}_{s''} \leq (\bar{r}_s - \underline{r}_{s'}) + (\bar{r}_{s'} - \underline{r}_{s''})$, and π_a is a star cost as in Appendix C-B; the remaining steps of that argument apply unchanged, so d^+_D is a pseudometric. Under the stated inclusions both refined operators lie between the original ones pointwise, which orders the fixed points. Finally, the proof of Theorem 6 uses only $d^+ \leq \max_a\{(1-\gamma)\bar{g}_a + \gamma\hat{w}_a(d^+) + \pi_a\}$ and $d^- \geq \max_a\{(1-\gamma)\underline{g}_a + \gamma\hat{w}_a(d^-) - \pi_a\}$, which hold here, so the same argument gives the stated source.

APPENDIX D
COMPLETE CLOSED-LOOP PROOF

The proof of Theorem 17 has three parts. A simulation argument shows that each episode succeeds with constant probability; a supermartingale bound converts this into a bound on the number of episodes; and the resolution of the metric then follows from the quota by the results of Sections III and IV.

A. Stopped Simulation Bound

Fix a policy and couple each true known-row transition maximally with its empirical counterpart. Until the first mismatch, the two systems share the same state, action, remaining horizon, and stopping rule. Frozen terminal priorities at unresolved rows lie in $[1/2, 1]$. A union bound over at most H transitions shows that the expected terminal payoffs of the two systems differ by at most $H\alpha_*$, uniformly over policies, because the row error is uniform. For any unresolved row, a true policy reaches it with probability at least p or stops earlier at another unresolved row, so its expected payoff is at least $p/2$. The empirical optimum is therefore at least $p/2 - H\alpha_*$, and the executed policy has true expected payoff at least $p/2 - 2H\alpha_* \geq p/4$. Its probability of sampling an unresolved row is at least this payoff.

B. An Adapted Success-Count Bound

Let $Q = SAn_0$ be the number of retained successes required and let $q = p/4$. For exploratory episode k, let $X_k = 1$ if an unresolved row is executed and $X_k = 0$ otherwise. To avoid conditioning on a future confidence event, we work with an auxiliary process that sets $X_k = 1$ after completion or after the first confidence failure. On every earlier good history, its conditional success probability is at least q, so for $\lambda = \log 2$,

$$\begin{aligned}\mathbb{E}[e^{-\lambda X_k} \mid \mathcal{H}_{k-1}] &\leq 1 - q + qe^{-\lambda} \\ &= 1 - q/2 \leq e^{-q/2}. \end{aligned} \tag{D.1}$$

Iterating conditional expectations and applying Markov's inequality,

$$\mathbb{P}\left(\sum_{k=1}^{N} X_k < Q\right) \leq \exp\{(\log 2)Q - pN/8\}. \tag{D.2}$$

With $N = \lceil 8(Q + \log(2/\delta))/p \rceil$ the right-hand side is at most $\delta/2$. On the model confidence event the auxiliary and actual processes agree until termination, and adding the failure probability $\delta_m = \delta/2$ of that event proves the bounds on episodes and interaction steps. If each reset costs at most c_{reset} physical steps, H is replaced by $H + c_{\mathrm{reset}}$ in the total.

C. Resolution, Compression, and Goal Policies

At termination every row holds exactly n_0 retained samples. By (41), $\beta^r \leq R_{\max}\alpha_*$ and $\beta^P \leq \alpha_*$, and (10) gives $b_a \leq 2R_{\max}\alpha_*$. Hence

$$\left\|d^+ - d^-\right\|_\infty \leq \frac{4R_{\max}\alpha_*}{1-\gamma} \leq \varepsilon/8. \tag{D.3}$$

Each computed endpoint is padded by a residual-plus-transport margin of at most $\varepsilon/16$, so the total widening is at most $\varepsilon/8$ and the reported width is at most $\varepsilon/4$. Path closure of the padded upper endpoint is still an upper bound and can only reduce the width; we call the resulting endpoints L and U. Applying Theorem 13 to L, U with $r = \varepsilon/2$, the separation scale is at least $\varepsilon/4$, so the number of centers is at most $\mathcal{N}_\sim(\varepsilon/8)$ and the distortion is at most ε. Every center has been visited, because every row has been sampled.

For a center g, plan the empirical finite-horizon hitting policy with terminal condition $V_0(s) = \mathbf{1}\{s = g\}$, treating g as absorbing for evaluation. The coupling argument over $H-1$ transitions gives a true success probability of at least $p - 2(H-1)\alpha_* \geq p/2$. This policy is learned; no goal reward is inferred from the extrinsic bisimulation distance. The statement about Euclidean fitting follows from Proposition 14 applied to the reported interval.

D. Exact Quotient Composition

For an exact homomorphism, the distribution of decoded successors and the mean reward are constant on each block, under every action. Decoded successor samples collected at different raw labels of one block therefore share a conditional transition law. The homomorphism fixes the reward means but not necessarily the full reward distributions; bounded rewards minus their common decoded mean form martingale differences, so the conditional Hoeffding bound replaces the independent-stream argument for pooled observations. The concentration of decoded successors and the union bound over visit counts are unchanged, and no common reward-noise distribution within a block is assumed.

Every raw transport coupling projects to a decoded coupling. Conversely, a decoded coupling lifts to a raw one by conditioning the raw successor distributions on each block, with the same cost for a block-constant ground metric. The lifted quotient metric is therefore the unique raw fixed point. The finite-state proof applies to the decoded rows and pulls back to raw states. A decoder learned in an earlier phase contributes its own confidence and sample cost, and a union bound proves Corollary 18.

APPENDIX E
COMPLETE LOWER-BOUND CONSTRUCTIONS

Both lower bounds use the same device. Absorbing self-loops make the metric an explicit function of the reward means, so a question about the geometry becomes a question about distinguishing Bernoulli parameters, for which change-of-measure arguments give sharp answers.

A. Geometry-Only Identification Obstruction

With absorbing self-loops, the transport cost for a pair equals its own ground cost, so a fixed point satisfies $d(i,j) = (1-\gamma)\max_a |r(i,a) - r(j,a)| + \gamma d(i,j)$, which proves (50). In the base instance all such distances are zero.

Let $N_{i,a}$ be the number of generative queries to row (i,a) before stopping, and let B_i be the event that the reported upper endpoint for $(i, \text{reference})$ is below 2ε. Under the base instance, valid intervals of width at most ε imply B_i, so $\mathbb{P}_0(B_i) \geq 1-\delta$. Under the alternative that changes (i,a), validity requires the upper endpoint to be at least 4ε, so $\mathbb{P}_{i,a}(B_i) \leq \delta$.

The chain rule for the relative entropy of an adaptively stopped experiment, followed by data processing onto B_i, gives

$$\begin{aligned}\mathbb{E}_0[N_{i,a}]\ \mathrm{kl}(1/2, 1/2+4\varepsilon) &\geq \mathrm{kl}(\mathbb{P}_0(B_i), \mathbb{P}_{i,a}(B_i)) \\ &\geq \mathrm{kl}(1-\delta, \delta), \end{aligned} \tag{E.1}$$

since all other row laws coincide between the two instances. For $\varepsilon \leq 1/16$,

$$\begin{aligned}\mathrm{kl}(1/2, 1/2+4\varepsilon) &= -\tfrac{1}{2}\log(1-64\varepsilon^2) \\ &\leq \frac{32\varepsilon^2}{1-64\varepsilon^2} \leq 64\varepsilon^2. \end{aligned} \tag{E.2}$$

Summing (E.1) over all SA nonreference rows gives the theorem. An infinite expected stopping time satisfies the bound trivially; otherwise the standard stopped change-of-measure inequality applies [1]. The base quotient has a single point, so its covering number is one and its doubling dimension is zero.

B. Controlled Two-Clock Separation

By the self-loop formula, the metric is the largest difference of action means. Between grid points, every nonzero coordinate difference is at least 8ε, while the unknown action contributes a difference of at most 4ε; this proves the grid part of (54). Differences in the known coordinates between the reference and a grid point are below $1/2$, whereas $\mu_q > 1/2$, so the unknown action determines every distance to the reference.

Grid labels are at least 8ε apart, and the reference is more than 2ε from every grid point. An ε-cover built from visited labels must therefore contain every label, and a generative schedule visits each one in $M+1$ queries. Each lattice ball is covered by at most 5^d lattice balls of half the radius. A ball that contains both the reference and a grid point has radius at least $\min_q \mu_q > 0.7$, whose half exceeds the grid diameter $1/4$, so two centers cover it. The doubling dimension is therefore $O(d)$, uniformly in m.

For identification, compare the instance with all $\theta_q = -1$ to the alternative that flips a single sign. An interval of width ε for that reference distance cannot be valid for both values, which are 4ε apart. As above, each unknown row requires an expected number of samples of at least

$$\frac{\mathrm{kl}(1-\delta, \delta)}{\mathrm{kl}(3/4 - 2\varepsilon, 3/4 + 2\varepsilon)}. \tag{E.3}$$

The inequality $\mathrm{kl}(u,v) \leq (u-v)^2/[v(1-v)]$ and $\varepsilon \leq 1/64$ bound the denominator by $128\varepsilon^2$, and summing over the M rows proves the lower bound.

For the upper bound, draw

$$n = \left\lceil \frac{8}{\varepsilon^2} \log \frac{2M}{\delta} \right\rceil \tag{E.4}$$

Bernoulli samples from each unknown row. By Hoeffding's inequality and a union bound, $|\hat{\mu}_q - \mu_q| \leq \beta = \sqrt{\log(2M/\delta)/(2n)} \leq \varepsilon/4$ simultaneously for all q. Report $[\hat{\mu}_q - \beta, \hat{\mu}_q + \beta]$, clipped to $[0,1]$, for each reference distance. The known means are exact, and even over these intervals any two unknown means differ by at most $4\varepsilon + 4\beta \leq 5\varepsilon < 8\varepsilon$. Grid-to-grid distances are therefore known exactly, and every reference interval has width at most $\varepsilon/2$. The total sample count is Mn plus one reference visit, which proves the upper bound. Since $m = 1/(32\varepsilon)$, $M = \Theta(\varepsilon^{-d})$.

APPENDIX F
NEURAL EXTENSION: TARGETS AND UPDATE ORDER

The guiding principle of this appendix is to change as little as possible when moving from finite labels to rich observations. Each finite-state quantity has a neural counterpart that is computed in the same way and plays the same role, so that any gap between the two can be traced to a specific approximation. We fix the targets, the parametrization, and

the update order of Algorithm F.1. The full specification is not evaluated in Section VIII, and we stress at the outset that ensemble disagreement, a shared encoder, and EMA targets do not by themselves give simultaneous model confidence or joint convergence.

A. Frozen Behavioral Targets

Let $z = f_{\bar\theta}(o)$. A masked bootstrap ensemble with randomized priors [2] predicts bounded mean rewards $\hat r_k(z,a)$ and stochastic successor laws $\hat P_k(\cdot \mid z,a)$. The likelihood models aleatoric randomness, and differences between heads serve as a proxy for epistemic uncertainty. We use the bounded, frozen ground cost

$$h_{\bar\theta}(o,o') = R_{\max}\min\{1, \|f_{\bar\theta}(o) - f_{\bar\theta}(o')\|_2\}. \tag{F.1}$$

With the same action at both observations, set

$$\begin{aligned} q_k(o,o',a) &= (1-\gamma)|\hat r_k(z,a) - \hat r_k(z',a)| \\ &\quad + \gamma\,\mathrm{OT}_{h_{\bar\theta}}(\hat P_k, \hat P_k'), \qquad \text{(F.2)} \\ M(o,o') &= \max_a K_{\mathrm{ens}}^{-1}\sum_k q_k(o,o',a), \\ \hat b_a(o,o') &= \beta_{\mathrm{cal}}\,\mathrm{sd}_k(q_k(o,o',a)) + \gamma\nu_{\mathrm{OT}}. \qquad \text{(F.3)} \end{aligned}$$

Bounded rewards and a bounded ground cost give $0 \le M \le R_{\max}$. The term ν_{OT} accounts for solver error, and β_{cal} is selected on a disjoint calibration split. Neither quantity substitutes for model coverage of unseen states. A calibration report must state the nominal level, the sampling unit, the sample size, and any distribution shift; marginal coverage on held-out data does not establish adaptive, simultaneous coverage.

For continuous actions, we fix a policy $\bar\pi$ within each epoch and use its induced reward and successor law. This defines a one-action policy metric $d^{\bar\pi}$, and the maximum over actions in the width critic below is dropped. A finite discretization of the actions instead defines a restricted-action metric. Neither construction equals the supremum over the continuous action set, and changing $\bar\pi$ changes the target metric.

B. A Bounded Width Critic and Explicit Endpoints

We use symmetric features $(|z-z'|, (z+z')/2, a)$ and an MLP g_ψ, with

$$Q_\psi^W(o,o',a) = R_{\max}\,\mathrm{sigmoid}(g_\psi), \qquad W_\psi = \max_a Q_\psi^W. \tag{F.4}$$

All self-pair outputs are set to zero. The parametrization enforces symmetry and boundedness, but neither the triangle inequality nor statistical validity. The surrogate endpoints, together with their midpoint and width, are

$$\begin{aligned} L_\psi &= \max\{0, M - W_\psi/2\}, \\ H_\psi &= \min\{R_{\max}, M + W_\psi/2\}, \\ \bar d_\psi &= (L_\psi + H_\psi)/2, \qquad w_\psi = H_\psi - L_\psi. \end{aligned} \tag{F.5}$$

Self-pair endpoints are zero. Note that W_ψ estimates a majorant, whereas w_ψ is the width of the clipped interval; the two differ. Before the first fit we use $M = R_{\max}/2$ and $W = R_{\max}$ off the diagonal, which gives $[L,H] = [0, R_{\max}]$.

On sampled successor supports, we optimize a feasible coupling $\hat\Lambda^a$ under the frozen lower endpoint L of the preceding epoch. With gradients stopped through every target quantity, the bounded target is

$$\begin{aligned} Y_a^W = \min\big\{&R_{\max}, 2\hat b_a + \gamma\xi_\Lambda \\ &+ \gamma\mathbb{E}_{(u,v)\sim\hat\Lambda^a} W_{\bar\psi}(u,v)\big\}. \end{aligned} \tag{F.6}$$

The critic minimizes $\mathbb{E}[(Q_\psi^W - Y_a^W)^2]$. The maximum over actions and the coupling expectation mirror Proposition 7; independent successors or a deterministic mean successor would define different targets. If the feasible coupling is suboptimal by ξ_Λ under the lower cost, then

$$\mathrm{OT}_{d^+}(\hat P, \hat P') - \mathrm{OT}_{d^-}(\hat P, \hat P') \le \langle \hat\Lambda^a, \Delta\rangle + \xi_\Lambda, \tag{F.7}$$

which is why the source in (F.6) contains $\gamma\xi_\Lambda$. When ξ_Λ cannot be bounded, the target is an empirical approximation. A certified critic would further require a uniform Bellman residual and valid model sources; a small squared error averaged over replay does not suffice.

C. Normalized Representation Learning

The encoder is fitted to either certified or surrogate endpoints, and which one is used must be recorded. Put $\tilde d = \bar d/R_{\max}$, $\tilde w = w/R_{\max}$, and $D_\theta = \|f_\theta(o) - f_\theta(o')\|_2$. With $\zeta > 0$,

$$\begin{aligned} \mathcal{L}_{\mathrm{rep}} = \tau_t\mathbb{E}_{\mathrm{temp}}&\frac{(D_\theta - \tilde d)^2}{\zeta^2 + \tilde w^2} \\ &+ \mathbb{E}_{\mathrm{strat}}\frac{(D_\theta - \tilde d)^2}{\zeta^2 + \tilde w^2}. \end{aligned} \tag{F.8}$$

Both terms regress onto the same frozen target, as required by (39). Stratified replay balances the cells that have been identified; it cannot balance states that have not. For EMA averages m_t and w_t of the normalized midpoint and width, set

$$\tau_t = \begin{cases} 1, & m_t + w_t = 0, \\ \mathrm{clip}(m_t/(m_t+w_t), \tau_{\min}, 1), & m_t + w_t > 0. \end{cases} \tag{F.9}$$

The temperature weights the temporal loss and never rescales a pairwise target. For fixed positive m_t it decreases as uncertainty grows, so the global term dominates while the geometry is uncertain.

Proposition F.1 (Reward-scale equivariance)**.** *Under $r \mapsto cr$ and $R_{\max} \mapsto cR_{\max}$ with $c > 0$, with reward radii scaled accordingly and transition radii unchanged, the exact endpoints, width, and source all scale by c. Their normalized versions, and hence* (F.8)*, are unchanged for the same normalized encoder.*

Proof. Transport is homogeneous in its ground cost, so $\mathcal{T}^{\pm}_{\mathrm{scaled}}[ch] = c\mathcal{T}^{\pm}[h]$, and uniqueness gives the scaled fixed points. Dividing by $cR_{\max}$ proves the claim. □

Identical optimizer trajectories additionally require consistently normalized outputs, regularizers, rewards, and solver tolerances.

Algorithm F.1 Neural EBG: specified extension, without an empirical certificate

Input: Encoder and targets, masked model ensemble, bounded pair critic, replay, anchors, calibration split, goal and task policies.

Set surrogate intervals to $[0, R_{\max}]$ off the diagonal.
for each target epoch **do**
  Refresh anchors; freeze the epoch policy and preceding lower proxy.
  Select a frontier by (F.10); navigate and explore.
  Fit bounded reward and stochastic successor heads with bootstrap masks.
  Freeze heads; form M, calibrated proxy sources, and feasible couplings.
  Update the width critic by (F.6).
  Freeze endpoints (F.5); fit (F.8).
  Update loss weight, policies, and EMA targets.
  Record calibration, failed navigation, model/solver error, and compute.
end for

D. Navigation, Training Order, and Scope

The agent keeps a memory C_t of observation anchors, refreshes their embeddings at each target epoch, and evaluates the novelty of a new observation before inserting it:

$$u_t(o) = \min_{c \in C_t} \{ D_{\bar{\theta}}(o, c) + \beta_u W_\psi(o, c)/R_{\max} \}. \tag{F.10}$$

This quantity plays the role of the optimistic novelty $\min_c d^+(s, c)$ in the finite-state construction. The agent selects a frontier goal, navigates to it with a goal-conditioned policy trained with hindsight relabeling [3], and then explores locally; returning before exploring follows the control pattern of Go-Explore [4]. The task policy may be trained on

$$r_t^{\text{train}} = r_t^{\text{ext}}/R_{\max} + \beta_t W_\psi(o_t, c_t)/R_{\max}, \tag{F.11}$$

where c_t is the nearest previous anchor. Because the bonus changes over time, it is not a policy-invariant potential reward, and it must be disabled when extrinsic performance is evaluated. SAC and PPO [5], [6] are compatible choices of policy learner; the finite-state study reports no result for either.

The main failure mode is collapse: unseen states can be mapped onto familiar embeddings, and then all heads agree, wrongly. The specification does not solve latent-state identification. A certified decoder would make the quotient composition of Corollary 18 available; without one, that guarantee does not transfer to the extension. Entropic transport error, solver brackets, and the computational cost of the extension are analyzed in Appendix H.

APPENDIX G
EXPERIMENTAL AND NUMERICAL SPECIFICATIONS

A. Sampling Design and Access Models

This appendix records what a reader would need to reproduce each study without the code: the seeds, the sampling schemes, the solver tolerances, and the statistical summaries. All configurations were fixed before execution, except the two post hoc diagnostics and the pilot-based designs named in Section VIII.

V1 uses 500 seeds per model, indexed 10000–10499. Independent, model-specific reward and successor streams generate nested binomial increments, which are exact sufficient statistics of the declared Bernoulli row laws. The count n is per state-action row, so the generative budget at count n is SAn samples, with no navigation. The two ring models are defined in (59) and Section VIII.

V2 uses seed indices 31000–31099. The dynamics and the current-row rewards are known, so their confidence radii are zero. Only the two future reward rows are sampled, which requires $2n$ Bernoulli observations at count n. Each seed uses the same future reward draws for all escape probabilities, and seeds are the independent units of aggregation. Local source, actual width, bounded pair majorant, global bound, and pair coverage are evaluated at every checkpoint.

V3 uses seed indices 51000–51031 on each task, with four matched priority and navigation variants. Common row streams give every variant the same first 64 reward and successor outcomes per row, and each planner sees only the observations it has acquired. Every allocated run contributes to the summary, with its completion status and interaction cost. The transitions of both tasks are given in Section VIII; for both, the Bernoulli rewards have means

$$\begin{aligned} r(i, 0) &= 0.05 + \frac{0.6i}{S-1}, \\ r(i, 1) &= 0.1 + \frac{0.7(3i \bmod S)}{S-1}. \end{aligned} \tag{G.1}$$

In the branching model, label $i < 3$ has child $2i+1+a$ under action $a \in \{0, 1\}$. Both tasks reset to label 0. The random controller samples actions uniformly. Optimistic navigation follows (44) and, as fixed in the original specification, breaks ties toward the smallest action index; Section VIII-G shows why the tie-breaking rule adopted in Section V is preferable. Source-only priorities use the local-source score with the normalization of (43), and propagated priorities use the same normalization with U_c. Priorities start at one and are refreshed every 128 episodes. An empirical transition row with no observations is initialized to a self-loop and its mean reward to zero; the maximal radii at zero count cover this initialization. An episode stops at its first unresolved-row sample or at its horizon and then resets. Steps taken through known rows count as interactions even when their observations are not retained. Resets are free. Goal policies are evaluated within $H - 1$ transitions, separately from the exploration horizon H.

V5 reuses the frozen V1 sufficient statistics at every checkpoint with $n \geq 128$, so all four interval methods see identical samples. The simulation-lemma interval is clipped to $[0, 1]$ off the diagonal, with ρ capped at one; it is valid on the event of Lemma 2 because $\left\| \mathcal{T}[h] - \hat{\mathcal{T}}[h] \right\|_\infty \leq \|b\|_\infty$ for every $h \in \mathfrak{C}$, where $\hat{\mathcal{T}}$ is the operator of the empirical model. For each seed and checkpoint, 200 bootstrap replicates redraw every row's reward successes and first-successor count from binomial laws with the empirical means, which is the nonparametric bootstrap for these Bernoulli and two-point row laws [7]. The pointwise interval takes the 5th and 95th percentiles of the

replicate metric at each pair. The simultaneous band takes the 90th percentile c of $\left\|\hat{d}^* - \hat{d}\right\|_\infty$ over replicates and reports $[\hat{d} - c, \hat{d} + c]$, clipped as above. Both have nominal level $0.9 = 1 - \delta$. Replicate seeds are derived from the V1 seed, the model, and the checkpoint.

V6 uses seed indices 80000–80099 for each $S \in \{16, 32, 64, 128\}$. Labels are split into four contiguous blocks of equal size. Block k has mean rewards $(0.1 + 0.25k,\ 0.85 - 0.25(3k \bmod 4))$; under action 0 it moves to block k with probability 0.7 and to block $k+1$ with probability 0.3, and under action 1 to block $k+2$ with probability 0.6 and to block $k+3$ with probability 0.4, indices modulo four. Each label draws its two successor labels uniformly from the designated blocks with a structure seed fixed by S, so every row has two successors and the model is exactly lumpable onto the four blocks. Rewards are Bernoulli. Nested prefixes of $2048, 8192, 32768, 131072$ samples per row are drawn as in V1. The anchor cover uses $r = 0.2$ at the final checkpoint, where every recorded width is below r. Certificate time is the median wall-clock time of five single-seed solves of $d^{\pm}$ and U_c at the final checkpoint, on one CPU process.

V7 uses seed indices 61000–61031 on the two V3 tasks with optimistic navigation and three priority rules: uniform ($v_t = 1$), source-only, and propagated. Every row must hold 4096 retained samples, priorities are refreshed every 512 episodes, and runs are capped at 10^6 steps. At the first episode boundary after 5000, 10000, 20000, 40000, and 80000 interaction steps, the snapshot endpoints are computed from all retained data with $\delta = 0.1$ and $\Gamma = 2$; a row with no samples has maximal radii. Common row streams are separate from those of V3 and are shared by the three rules. The quota and the budget grid were set after a pilot with seed 61000 on two task–rule pairs and a quota of 256, in which the recorded widths stayed at or near one; a timing check of the final configuration on the same seed preceded the full run. The pilot runs are not part of the reported summaries. The specified planner breaks ties toward the smallest action index. The post hoc diagnostic of Section VIII-G reruns the same seeds, tasks, and rules with ties broken toward executing an unresolved row, using a tolerance of 10^{-12}; its specification was written after the specified V7 results had been analyzed and is stored separately.

V8 regenerates the V6 statistics from the same seeds, which reproduces the V6 samples exactly. The KL intervals use the budget ℓ_n of (6); the diameter-refined certificate keeps Weissman radii with the declared Γ, whereas the support-refined one uses each row's own support size. Refinement stops when an iteration changes the padded upper endpoint by less than 10^{-9} or after 40 iterations. A pair counts as certified distinct when $d^- > 10^{-9}$. For $S \in \{16, 32\}$, a simultaneous bootstrap band is formed as in V5.

V9 uses the six-state RiverSwim variant of Osband et al. [8], introduced by Strehl and Littman [9]. The left action is deterministic. The right action stays at label 0 with probability 0.4 and otherwise advances; at interior labels it moves left, stays, or advances with probabilities 0.05, 0.6, and 0.35; at the last label it moves left with probability 0.4 and otherwise stays. Bernoulli rewards have mean 0.005 for the left action at label 0 and one for the right action at label 5, and $\gamma = 0.9$. Generative runs use seeds 90000–90099 with multinomial successor counts, $\Gamma = 3$, and a bootstrap band built from multinomial and binomial resampling. The navigation runs use seeds 92100–92131.

V10 uses seeds 63100–63131 for the full certificate, with budgets of 20000 to 320000 steps and $H = 20$ on RiverSwim. The source-only score is the local source of the width recursion in Proposition 12, $(1-\gamma)(\bar{g}_a - \underline{g}_a) + 2\pi_a$ with baseline radii, which equals $2b_a$ of (10) unless the observed reward gap is smaller than e_a^r; V7 used $2b_a$, so the two studies are not compared directly. The targeted query uses seeds 64100–64131 and stops when the baseline lower endpoint of the target pair exceeds zero. Its planner may sample the current row with payoff v_t or, once the row holds 16 samples, cross it under the empirical law, and every executed transition is retained. The count rule scores a row by its own radius term $(1-\gamma)\beta^r + \gamma R_{\max}\beta^P$. The adjoint rule uses (47) with all weight on the target pair, retires a row with zero influence once it holds 16 samples, and breaks ties by a 10^{-3} multiple of the radius term. Both map scores to $[1/2, 1]$. Pilot runs on seeds 63000, 64000, 64001, and 92000 fixed these designs and are not reported.

V11 uses seeds 70000–70004 per method. The encoder and critic are two-layer SiLU networks with 64 hidden units and an eight-dimensional embedding, trained with Adam at learning rate 10^{-3} on 64 pairs per update, with target networks averaged at rate 0.01, $\beta_{\mathrm{cal}} = 2$, and $\zeta = 0.05$. Successor sets of eight observations are matched by exact assignment, and each of five bootstrap heads resamples them with replacement. Evaluation uses 300 held-out observations per seed. The bias diagnostic draws 3000 random latent pairs per successor count with seed 71000.

B. Budgets of the Two-Clock Family

Table G.1 lists the budgets that V4 evaluates from the formulas of Theorem 20.

C. Solvers and Error Accounting

All arrays are float64. On rows with at most two successors, the feasible 2×2 coupling is parametrized by

$$z \in [\max\{0, p_0 + q_0 - 1\}, \min\{p_0, q_0\}]. \tag{G.2}$$

The transport objective is affine in z, so the minimizing endpoint is determined by the sign of $h_{00} - h_{01} - h_{10} + h_{11}$. Singleton supports are handled as cases with zero mass on one successor. Four coefficients apply the pair kernel without forming an AS^4 tensor. Independent checks compare this calculation with scalar primal transport and with linear programming on supported examples.

Fixed-point iteration stops when the step norm divided by $1 - \gamma$ is at most 2×10^{-11}, after which the actual operator residual is evaluated and endpoint padding is set to that residual divided by $1-\gamma$. The possible suboptimality of the plan for the lower cost is bounded by twice the error of the lower fixed point, and the width majorant receives its own residual margin plus the endpoint padding. Coverage, midpoint, triangle, and

TABLE G.1
V4: CALCULATED BUDGETS FROM THE CONTROLLED-FAMILY THEOREM, WITH $M = m^d$, $\varepsilon = 1/(32m)$, AND $n = \lceil 8\varepsilon^{-2}\log(2M/0.1)\rceil$ SAMPLES PER ROW. THE COVER BUDGET $M+1$ GROWS AS m^d; THE PRESCRIBED RESOLUTION BUDGET $Mn+1$ GROWS AS m^{d+2} UP TO THE LOGARITHM IN n. THESE ARE FORMULA EVALUATIONS, NOT OBSERVED STOPPING TIMES OR FITTED SCALING LAWS, AND CARRY NO CONFIDENCE BANDS; $Mn+1$ IS ROUNDED TO THREE SIGNIFICANT DIGITS.

		$d=1$			$d=2$			$d=3$		
m	ε^{-1}	$M+1$	n	$Mn+1$	$M+1$	n	$Mn+1$	$M+1$	n	$Mn+1$
4	128	5	574361	2.30×10^{6}	17	756066	1.21×10^{7}	65	937770	6.00×10^{7}
6	192	7	1411889	8.47×10^{6}	37	1940301	6.99×10^{7}	217	2468712	5.33×10^{8}
8	256	9	2660853	2.13×10^{7}	65	3751079	2.40×10^{8}	513	4841306	2.48×10^{9}
10	320	11	4340382	4.34×10^{7}	101	6226660	6.23×10^{8}	1001	8112938	8.11×10^{9}
12	384	13	6465225	7.76×10^{7}	145	9396540	1.35×10^{9}	1729	12327856	2.13×10^{10}

majorant checks use the tolerance 2×10^{-8}. These are floating-point checks, not interval-arithmetic proofs that account for machine rounding.

At the final low-discount V1 checkpoint, the anchor audit builds the greedy cover of the upper metric with radius 0.25 and evaluates, for every pair, the infinity-norm representation distance, the covering radius, and the distortion. With all labels as anchors, it also checks that the representation distance equals the upper metric.

D. Statistical Summaries and Failed Runs

For calibration we report the all-pair indicator at each count and its conjunction over the seven checkpoints. Exact two-sided Clopper–Pearson intervals describe the probability of run coverage for each fixed model; they do not estimate coverage conditional on the unobservable good event. Means and their uncertainty are computed across independent sampling seeds, not across correlated state pairs or nested checkpoints.

Mean widths, errors, propagation diagnostics, and ablation costs use 5000 seed-bootstrap resamples with analysis seed 899, and the 2.5th and 97.5th percentiles give the reported 95% intervals. Priority contrasts resample matched differences between seeds. The intervals are descriptive; we make no familywise significance claim across contrasts.

All allocated seeds remain in every summary. A run that does not complete within its step cap contributes the capped cost, so the endpoint is an acquisition cost under a restricted budget; every interaction run reported in this paper completed.

APPENDIX H
TRANSPORT ERROR AND COMPUTATION IN THE NEURAL EXTENSION

The neural extension of Appendix F replaces exact transport by entropic solvers on sampled supports. This appendix bounds the error that the replacement introduces and records what the extension costs to compute. For finite supports, entropy-regularized transport is

$$\mathrm{OT}_h^{\lambda}(p,q) = \min_{\Lambda\in\Pi(p,q)} \{\langle \Lambda, h\rangle + \lambda\,\mathrm{KL}(\Lambda\|p\otimes q)\}. \tag{H.1}$$

Sinkhorn iterations [10], [11] approximate this objective, whose bias satisfies

$$0 \le \mathrm{OT}_h^{\lambda} - \mathrm{OT}_h \le \lambda \min\{H(p), H(q)\}. \tag{H.2}$$

The upper bound follows by evaluating an unregularized optimal coupling, whose mutual information is at most the entropy of either marginal. Feasible primal plans and unregularized dual potentials together give a direct numerical bracket. Debiased Sinkhorn divergences [12], [13] need not preserve the γ-contraction when the ground cost varies; a direct three-term bound has Lipschitz factor at most two.

Dense neural couplings cost $O(B_{\mathrm{pair}}A_{\mathrm{eval}}K_{\mathrm{ens}}m^2n_{\mathrm{S}})$ for m successor samples and n_{S} Sinkhorn iterations, and shared plans need their own error checks. Finite sparse plans can instead be stored in $O(AS^2\Gamma^2)$ space; with two successors per row, the finite-state implementation applies four coefficients per plan without forming a dense AS^4 transition tensor. Resource comparisons must include model fitting, calibration, anchor refresh, and navigation.

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